

ELITE MATHEMATICS

Pearson Edexcel International A Level

Mechanics 1

Adaptive Classified

Chapter 10 • Moments

With Answers

Questions with full worked solutions

- 42 pages
- Past papers up to June 2026

MECHANICS 1 | TOPIC 10

Moments

Your classified route

19

QUESTIONS

147

MARKS

3

FAMILIES

9

VARIANTS

DR ESLAM AHMED

CLASSIFIED BY IDEA

Read the diagram, choose the model, then use the working space to complete the mechanics.

TOPIC 10 | FAMILY 01

Beam/rod equilibrium and support forces

WHAT THIS FAMILY ASKS

Combine moments and vertical equilibrium to find reactions, tensions or attached loads.

VARIANT ROUTE**V01** Two supports or strings**V02** Given reaction/tension ratio**V03** Attached particle or package**V04** Support or rope capacity limits an attached load**V05** Admissible range from non-negative reactions**10 classified questions**

83 total marks

January 2026 Q4 — complete original question

4.

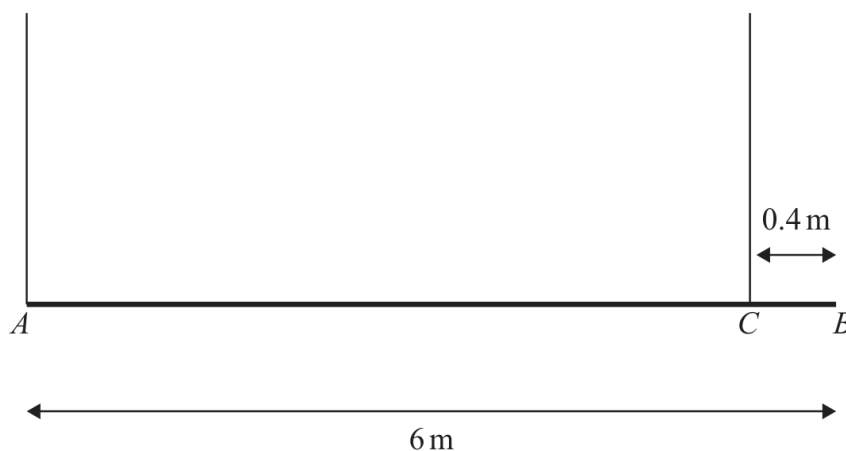


Figure 2

Figure 2 shows a uniform rod AB of length 6 m suspended by two light vertical ropes.

The first rope is attached to the rod at A .

The second rope is attached to the rod at the point C , where $CB = 0.4$ m.

The rod has mass 30 kg.

A particle of mass 20 kg is now attached to the rod at B .

The rod is in equilibrium in a horizontal position.

(a) Find

- (i) the tension in the rope attached to the rod at C ,
- (ii) the tension in the rope attached to the rod at A .

(6)

The particle of mass 20 kg at B is removed and replaced by a particle of mass M kg.

The rod remains in equilibrium in a horizontal position.

(b) Find the exact maximum value of M .

(3)

January 2026 Q4 — detailed solution

Two supports

ALL
9 marks**Set the distances**

The rod is 6 m long and C is 0.4 m from B, so $AC = 5.6$ m. The rod is uniform, so its weight $30g$ acts at the midpoint, 3 m from A. The 20 kg particle at B has weight $20g$.

Take moments about A

Moments about A remove the tension at A. Clockwise moments from the rod and particle equal the anticlockwise moment from the rope at C: $T_C(5.6) = 30g(3) + 20g(6)$.

Find the tension at C

$T_C(5.6) = 210g$, so $T_C = 37.5g = 367.5$ N. To suitable accuracy, $T_C = 368$ N, or 370 N to 2 significant figures.

Use vertical equilibrium

Vertically, $T_A + T_C = 30g + 20g = 50g$. Therefore $T_A = 50g - 37.5g = 12.5g = 122.5$ N, so $T_A = 123$ N, or 120 N to 2 significant figures.

Use the limiting case for maximum M

For the greatest possible mass at B, the rope at A is just slack, so $T_A = 0$. Take moments about C: the clockwise moment of Mg at B balances the anticlockwise moment of the rod weight.

Solve for M

The distance from C to B is 0.4 m and the rod weight acts $5.6 - 3 = 2.6$ m to the left of C. Hence $Mg(0.4) = 30g(2.6)$, so $M = \frac{78}{0.4} = 195$.

Final answer (a)(i) $T_C = 368$ N or 370 N. (a)(ii) $T_A = 123$ N or 120 N. (b) $M = 195$.

January 2020 Q2 — complete original question

2.

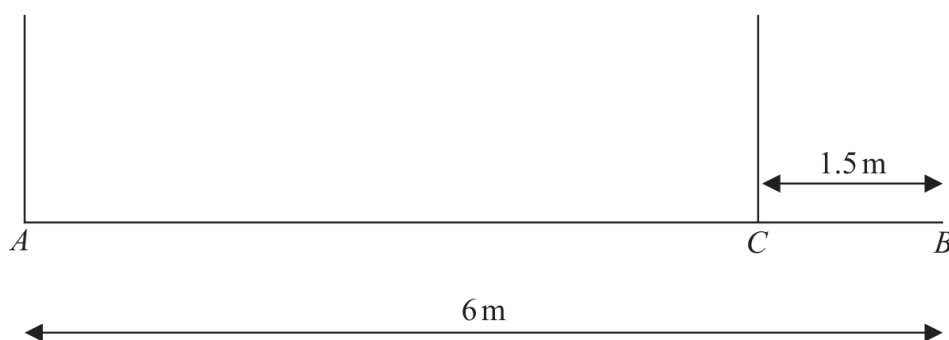


Figure 1

A non-uniform beam AB has length 6 m and weight W newtons. The beam is supported in equilibrium in a horizontal position by two vertical ropes, one attached to the beam at A and the other attached to the beam at C , where $CB = 1.5$ m, as shown in Figure 1.

The centre of mass of the beam is 2.625 m from A .

The ropes are modelled as light strings. The beam is modelled as a non-uniform rod.

Given that the tension in the rope attached at C is 20 N greater than the tension in the rope attached at A ,

(a) find the value of W .

(6)

(b) State how you have used the fact that the beam is modelled as a rod.

(1)

January 2020 Q2 — detailed solution

Given reaction ratio

ALL
7 marks**Set the tensions**

Let the tension at A be T . The tension at C is then $T + 20$. Since $CB = 1.5$ m and $AB = 6$ m, $AC = 4.5$ m.

Use vertical equilibrium

The upward tensions balance the weight: $T + (T + 20) = W$, so $2T + 20 = W$.

Take moments about A

The centre of mass is 2.625 m from A. Taking moments about A gives $4.5(T + 20) = 2.625W$.

Solve for W

From the moments equation, $T + 20 = \frac{2.625}{4.5}W = \frac{7}{12}W$. Also $T + 20 = \frac{W + 20}{2}$. Hence $\frac{W + 20}{2} = \frac{7W}{12}$, so $6W + 120 = 7W$ and $W = 120$ N.

State the modelling use

Modelling the beam as a rod means it stays straight and rigid, so the distances used in the moments equations do not change.

Final answer (a) $W = 120$ N. (b) The rod model keeps the beam straight and rigid, so the stated distances along the beam are fixed.

Summer 2023 Q4 — complete original question

4.

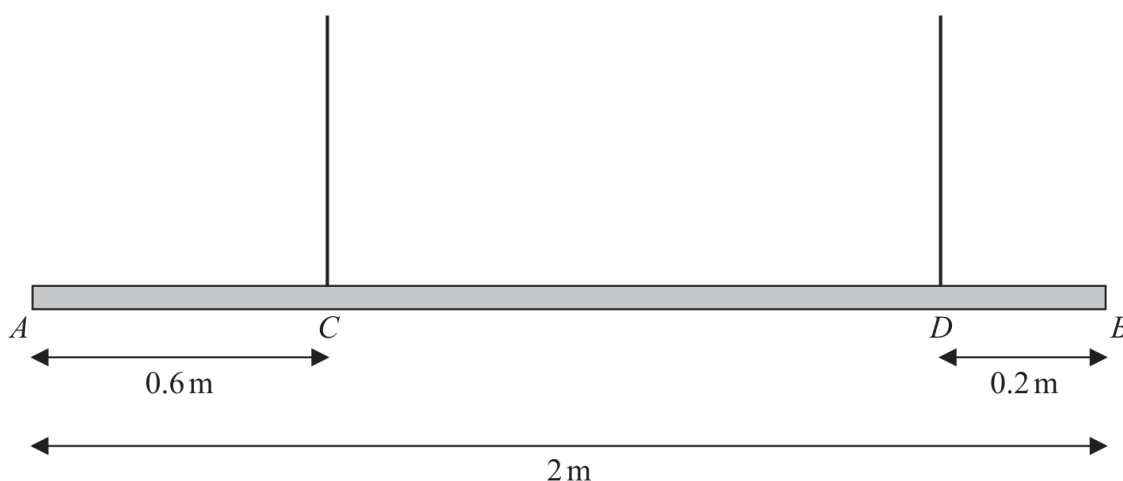


Figure 1

Figure 1 shows a beam AB , of mass m kg and length 2 m, suspended by two light vertical ropes.

The ropes are attached to the points C and D on the beam, where $AC = 0.6$ m and $DB = 0.2$ m

The beam is in equilibrium in a horizontal position.

A particle of mass pm kg is attached to the beam at A and the beam remains in equilibrium in a horizontal position.

The beam is modelled as a uniform rod.

- (a) Given that the tension in the rope attached at C is four times the tension in the rope attached at D , use the model to find the exact value of p .

(7)

The particle of mass pm kg at A is removed and replaced by a particle of mass qm kg at A .

The beam remains in equilibrium in a horizontal position but is now on the point of tilting.

- (b) Using the model, find the exact value of q

(4)

- (c) State how you have used the modelling assumption that the beam is uniform.

(1)

Summer 2023 Q4 — detailed solution

Given reaction ratio

ALL
12 marks

Set the geometry

The beam is 2 m long. Since $AC = 0.6$ m and $DB = 0.2$ m, point D is 1.8 m from A. The uniform beam has weight mg acting 1.0 m from A.

Name the tensions

Let the tension at D be T . The tension at C is four times this, so it is $4T$.

Use vertical equilibrium

$4T + T = pmg + mg$, so $5T = (p + 1)mg$.

Take moments about A

The particle at A has no moment about A. Hence $4T(0.6) + T(1.8) = mg(1.0)$.

Find T

$2.4T + 1.8T = mg$, so $4.2T = mg$ and $T = \frac{5mg}{21}$.

Find p

Substitute into $5T = (p + 1)mg$: $5 \cdot \frac{5mg}{21} = (p + 1)mg$. Thus $p + 1 = \frac{25}{21}$ and $p = \frac{4}{21}$.

Use the tilting condition

When the particle of mass qm is at A and the beam is on the point of tilting, the rope at D is just slack, so its tension is zero.

Take moments about C

About C, the particle at A has moment $qmg(0.6)$ and the beam weight has moment $mg(0.4)$ in the opposite direction. Thus $qmg(0.6) = mg(0.4)$, giving $q = \frac{2}{3}$.

Use the modelling assumption

The beam is uniform, so its centre of mass is at the midpoint of the beam and its weight acts there.

Final answer (a) $p = \frac{4}{21}$. (b) $q = \frac{2}{3}$. (c) The beam weight acts at its midpoint.

October 2023 Q1 — complete original question

1.

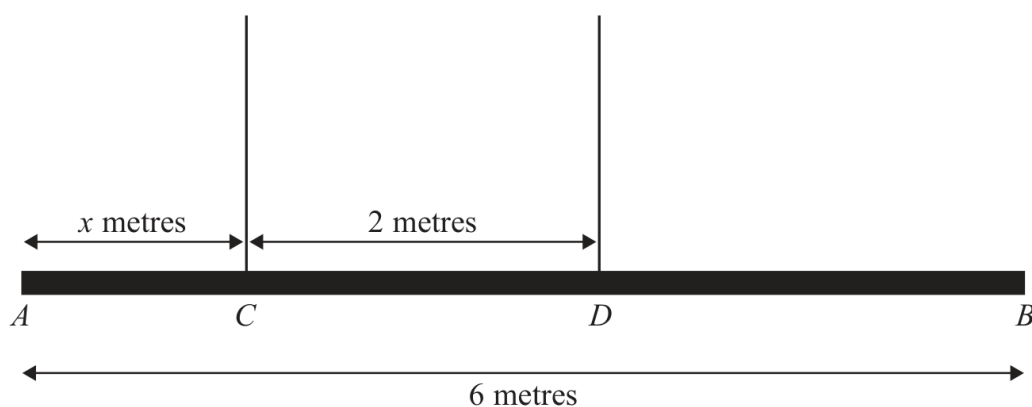


Figure 1

Figure 1 shows a beam AB with weight 24 N and length 6 m .

The beam is suspended by two light vertical ropes. The ropes are attached to the points C and D on the beam where $AC = x$ metres and $CD = 2\text{ m}$.

The tension in the rope attached to the beam at C is double the tension in the rope attached to the beam at D .

The beam is modelled as a uniform rod, resting horizontally in equilibrium.

Find

- (i) the tension in the rope attached to the beam at D .
- (ii) the value of x .

(5)

October 2023 Q1 — detailed solution

Given reaction ratio

ALL
5 marks**Name the tensions**

Let the tension at D be T . The tension at C is double this, so it is $2T$.

Use vertical equilibrium

The beam is in equilibrium, so the upward forces equal the weight: $2T + T = 24$. Hence $3T = 24$ and $T = 8$ N.

Set the moment distances

The beam is uniform, so its weight acts at the midpoint, 3 m from A. Also $AC = x$ and $AD = x + 2$.

Take moments about A

Clockwise and anticlockwise moments balance: $(2T)x + T(x + 2) = 24 \cdot 3$.

Solve for x

Substitute $T = 8$: $16x + 8(x + 2) = 72$. Thus $24x + 16 = 72$, so $24x = 56$ and $x = \frac{7}{3}$ m.

Final answer (i) $T_D = 8$ N. (ii) $x = \frac{7}{3}$ m ≈ 2.33 m.

January 2024 Q5 — complete original question

5.

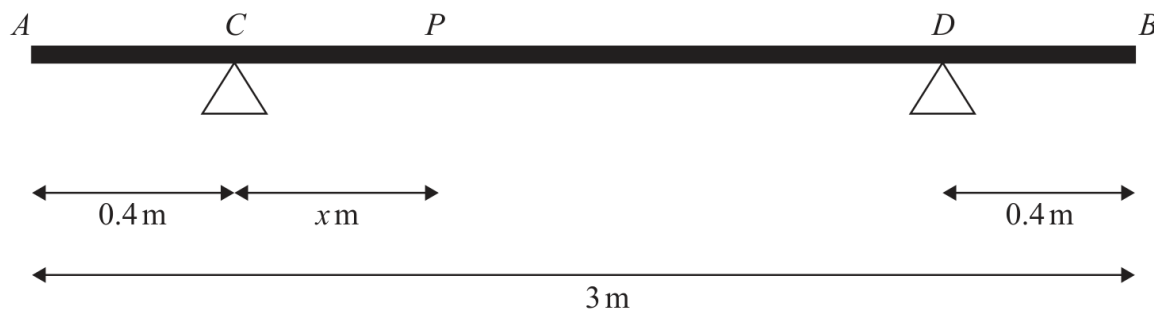


Figure 4

A beam AB has mass 30 kg and length 3 m.

The beam rests on supports at C and D where $AC = 0.4$ m and $DB = 0.4$ m, as shown in Figure 4.

A person of mass 55 kg stands on the beam between C and D .

The person is modelled as a particle at the point P , where $CP = x$ metres and $0 < x < 2.2$

The beam is modelled as a uniform rod resting in equilibrium in a horizontal position.

Using the model,

- (a) show that the magnitude of the reaction at C is $(686 - 245x)$ N. (3)

The magnitude of the reaction at C is **four** times the magnitude of the reaction at D .

Using the model,

- (b) find the value of x (4)

The person steps off the beam and places a package of mass M kg at A .

The package is modelled as a particle at the point A .

The beam is now on the point of tilting about C .

Using the model,

- (c) find the value of M (3)

January 2024 Q5 — detailed solution

Given reaction ratio

ALL
10 marks**Set the positions on the beam**

$AC = 0.4$ m and $DB = 0.4$ m, so $CD = 2.2$ m. The beam is uniform, so its weight $30g$ acts at the midpoint, which is 1.1 m from D. The person is $2.2 - x$ m from D.

Take moments about D

Taking moments about D removes R_D : $R_C(2.2) = 55g(2.2 - x) + 30g(1.1)$.

Show the expression for R_C

Using $g = 9.8$, $R_C = \frac{539(2.2 - x) + 294(1.1)}{2.2} = 686 - 245x$.

Use the reaction ratio

Given $R_C = 4R_D$ and vertical equilibrium gives $R_C + R_D = (55 + 30)g = 833$. Hence $5R_D = 833$ and $R_C = 666.4$.

Find x

$686 - 245x = 666.4$, so $245x = 19.6$ and $x = 0.08$.

Use the tilting condition with the package

When the package is at A and the beam is on the point of tilting about C, the reaction at D is zero.

Find M

Take moments about C: $Mg(0.4) = 30g(1.1)$. Thus $M = \frac{33}{0.4} = 82.5$.

Final answer (a) $R_C = 686 - 245x$. (b) $x = 0.08$. (c) $M = 82.5$.

October 2019 Q5 — complete original question

5.

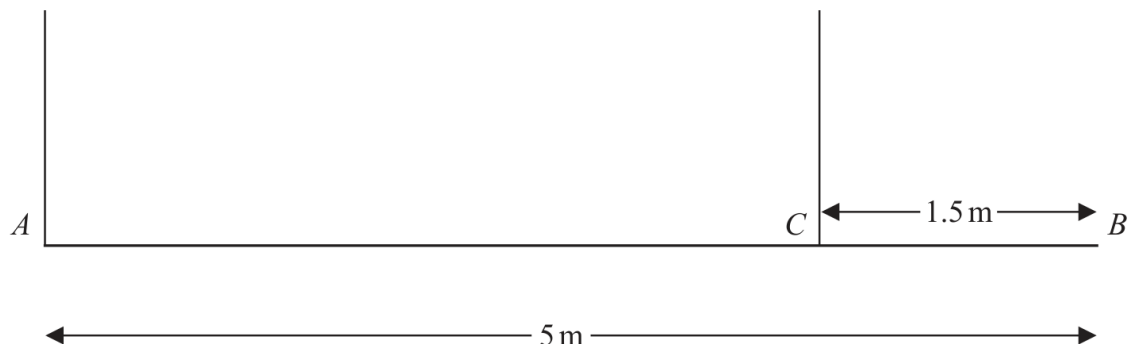


Figure 2

A non-uniform beam, AB , has length 5 m and mass 12 kg. The beam is suspended in a horizontal position by two vertical ropes. One rope is attached to the beam at A . The other rope is attached to the beam at C , where $CB = 1.5$ m, as shown in Figure 2. The distance of the centre of mass of the beam from A is 1.75 m. The beam is modelled as a non-uniform rod and the ropes are modelled as light inextensible strings.

A particle of mass M kg is now placed on the beam at B and the beam remains in equilibrium in a horizontal position.

(a) Find the largest possible value of M .

(3)

The particle at B is now removed and a particle of mass 15 kg is now placed on the beam at the point D , where $AD = x$ metres. The beam remains in equilibrium in a horizontal position.

Given that the tension in the rope attached to the beam at C is now twice the tension in the rope attached to the beam at A ,

(b) find the value of x .

(5)

October 2019 Q5 — detailed solution

Attached load

ALL
8 marks**Locate the key distances**

The beam is 5 m long. Since $CB = 1.5$ m, $AC = 3.5$ m. The centre of mass is 1.75 m from A, so it is also 1.75 m from C.

Use the largest M condition

For the largest possible M at B, the rope at A is just slack, so the beam is supported at C only.

Take moments about C

The particle at B is 1.5 m to the right of C, and the beam's weight acts 1.75 m to the left of C. Hence $Mg(1.5) = 12g(1.75)$.

Find M

$$M = \frac{12(1.75)}{1.5} = 14.$$

Set the tensions for part b

Let the tension at A be T . The tension at C is $2T$. Vertical equilibrium gives $T + 2T = 12g + 15g = 27g$, so $T = 9g$ and the tension at C is $18g$.

Take moments about A

For the new particle at $AD = x$, moments about A give $18g(3.5) = 12g(1.75) + 15gx$.

Solve for x

Cancel g : $63 = 21 + 15x$. Thus $15x = 42$ and $x = 2.8$ m.

Final answer (a) $M = 14$. (b) $x = 2.8$ m.

October 2022 Q2 — complete original question

2.

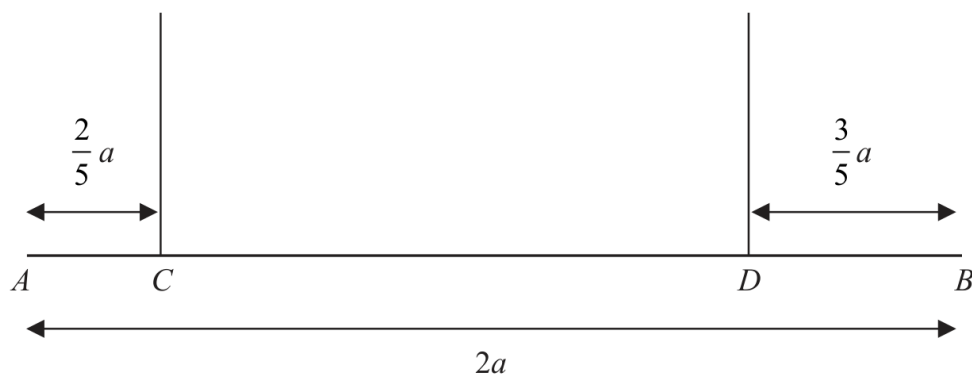


Figure 1

A uniform rod AB has length $2a$ and mass M . The rod is held in equilibrium in a horizontal position by two vertical light strings which are attached to the rod at C and D ,

where $AC = \frac{2}{5}a$ and $DB = \frac{3}{5}a$, as shown in Figure 1.

A particle P is placed on the rod at B .

The rod remains horizontal and in equilibrium.

(a) Find, in terms of M , the largest possible mass of the particle P (3)

Given that the mass of P is $\frac{1}{2}M$

(b) find, in terms of M and g , the tension in the string that is attached to the rod at C . (3)

October 2022 Q2 — detailed solution

Attached load

ALL
6 marks**Set the distances**

The rod length is $2a$. The centre of mass is at the midpoint, distance a from A. Also $AC = \frac{2a}{5}$ and $AD = 2a - \frac{3a}{5} = \frac{7a}{5}$.

Use the largest mass condition

For the largest possible mass at B, the string at C is just slack and the rod is about to tilt about D.

Take moments about D

Let the particle mass be X . The distance from D to B is $\frac{3a}{5}$ and the distance from D to the rod's centre of mass is $\frac{2a}{5}$. Hence $Xg \left(\frac{3a}{5} \right) = Mg \left(\frac{2a}{5} \right)$.

Find the largest mass

Cancel $ga/5$: $3X = 2M$, so $X = \frac{2M}{3}$.

Use the given mass

Now the particle has mass $\frac{1}{2}M$. Let the tensions at C and D be T_C and T_D .

Take moments about D

About D, $T_C(a) + \frac{1}{2}Mg \left(\frac{3a}{5} \right) = Mg \left(\frac{2a}{5} \right)$.

Find the tension at C

$T_C a + \frac{3}{10}Mga = \frac{2}{5}Mga$, so $T_C = \frac{1}{10}Mg$.

Final answer (a) largest mass = $\frac{2M}{3}$. (b) $T_C = \frac{1}{10}Mg$.

January 2025 Q4 — complete original question

4.

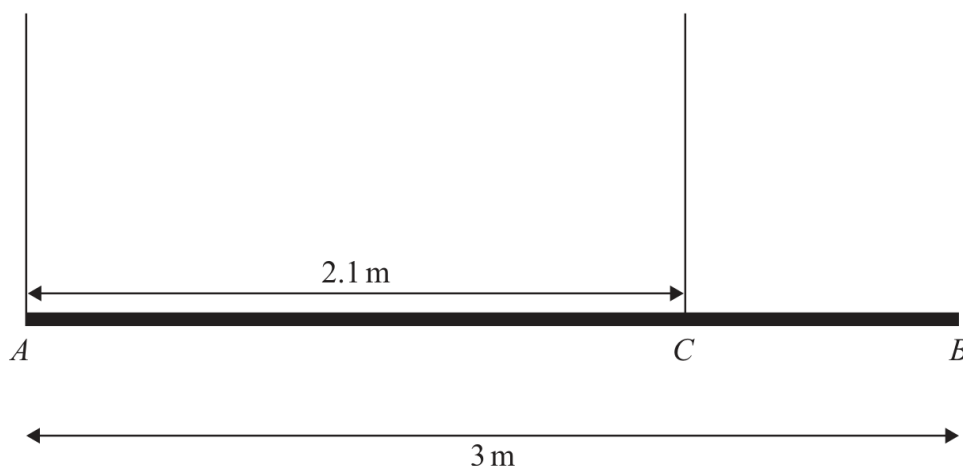


Figure 2

A uniform rod AB has length 3 m and weight W newtons.
 The rod is suspended by two light vertical ropes.
 One rope is attached to the rod at A and the other rope is attached to the rod at C ,
 where $AC = 2.1$ m.

The rod is in equilibrium in a horizontal position, as shown in Figure 2.

The tension in the rope at C is 350 N.

(a) Show that $W = 490$

(3)

A particle P of weight 210 N is attached to the rod at a distance d metres from A .
 The tension in the rope at C is now 600 N.
 The rod remains in equilibrium in a horizontal position.

(b) Find the value of d .

(3)

Particle P is removed from the rod.

A particle Q of weight X newtons is now attached at B .

The rod remains in equilibrium in a horizontal position and is now on the point of tilting.

(c) Find the value of X .

(4)

January 2025 Q4 — detailed solution

Attached load

ALL
10 marks**Use moments to find W**

Take moments about A. The rope at C is 2.1 m from A and the weight of the uniform rod acts at its midpoint, 1.5 m from A. Hence $350(2.1) = W(1.5)$.

Show W is 490

$735 = 1.5W$, so $W = 490$ N.

Set up moments with particle P

When P is attached, the tension at C is 600 N. Taking moments about A: $600(2.1) = 490(1.5) + 210d$.

Find d

$1260 = 735 + 210d$, so $525 = 210d$ and $d = 2.5$.

Use the tilting condition

When Q is attached at B and the rod is on the point of tilting, the rope at A is just slack. The rod is about to tilt about C.

Find X

Take moments about C. The rod weight acts $2.1 - 1.5 = 0.6$ m to the left of C, and Q acts $3 - 2.1 = 0.9$ m to the right. Thus $490(0.6) = X(0.9)$, so $X = \frac{980}{3}$ N.

Final answer (a) $W = 490$. (b) $d = 2.5$. (c) $X = \frac{980}{3}$ N ≈ 327 N.

October 2025 Q1 — complete original question

1.

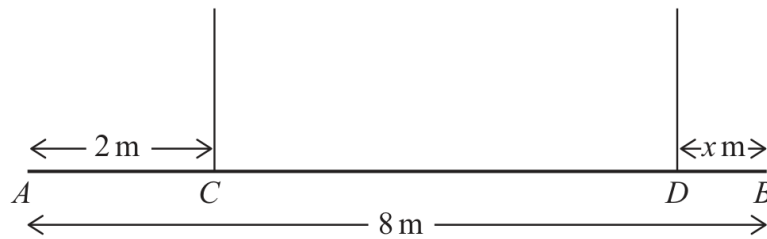


Figure 1

Figure 1 shows a sketch of a beam AB , with weight 240 N and length 8 m .

The beam is held in equilibrium in a horizontal position by two vertical ropes. The ropes are attached to the beam at the points C and D , where $AC = 2\text{ m}$ and $DB = x$ metres.

The beam is modelled as a uniform rod and the ropes are modelled as light inextensible strings.

The tension in the rope at D is 90 N .

(a) Show that $x = \frac{2}{3}$

(3)

The rope at C will break if its tension exceeds 183 N . The rope at D cannot break. A package of weight W newtons is now attached to the beam at A . The beam remains horizontal and in equilibrium.

The package is modelled as a particle.

It is given that the rope at C does not break.

(b) Find the greatest possible value of W .

(4)

October 2025 Q1 — detailed solution

Support or rope capacity limit

ALL
7 marks**Part (a): Show that $x = \frac{2}{3}$ - Identify distances from C**

From the diagram:

; Distance AC = 2 m ; Distance CD = 8 - 2 - x = (6-x) m ; Distance from A to center = 4 m, so distance from C to center = 4 - 2 = 2 m

Part (a): Show that $x = \frac{2}{3}$ - Take moments about C

$$\sum M_C = 0$$

Clockwise moments = Anticlockwise moments

$$90 \times (6 - x) = 240 \times 2 \quad 90(6 - x) = 480 \quad 540 - 90x = 480 \quad 90x = 60 \quad x = \frac{60}{90} = \frac{2}{3}$$

Part (b): Find the greatest possible value of W - Apply vertical equilibrium

$$\sum F_{\text{vertical}} = 0$$

$$T_C + T_D = 240 + W \quad 183 + T_D = 240 + W \quad T_D = 57 + W$$

Part (b): Find the greatest possible value of W - Take moments about C

$$\sum M_C = 0$$

Clockwise moments = Anticlockwise moments

$$\text{Distance CD} = 6 - \frac{2}{3} = \frac{16}{3} \text{ m}$$

$$T_D \times \frac{16}{3} + W \times 2 = 240 \times 2 \quad (57 + W) \times \frac{16}{3} + 2W = 480 \quad \frac{16(57+W)}{3} + 2W = 480 \quad \frac{912+16W}{3} + 2W = 480$$

$$\text{Multiply through by 3: } 912 + 16W + 6W = 1440 \quad 22W = 528 \quad W = 24$$

Final answer (a) $x = \frac{2}{3}$. (b) $W = 24 \text{ N}$.

Summer 2022 Q5 — complete original question

5.

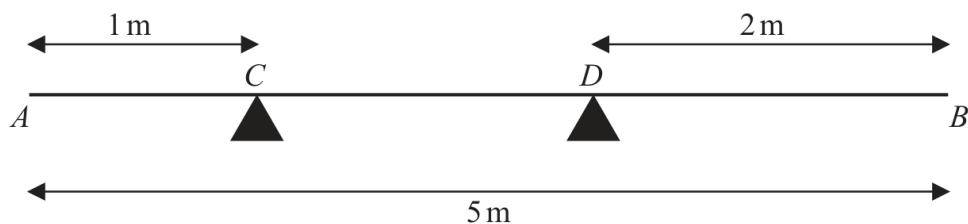


Figure 2

A uniform rod AB has length 5 m and mass 5 kg. The rod rests in equilibrium in a horizontal position on two supports C and D , where $AC = 1$ m and $DB = 2$ m, as shown in Figure 2.

A particle of mass 10 kg is placed on the rod at A and a particle of mass M kg is placed on the rod at B . The rod remains horizontal and in equilibrium.

- (a) Find, in terms of M , the magnitude of the reaction on the rod at C . (3)
- (b) Find, in terms of M , the magnitude of the reaction on the rod at D . (3)
- (c) Hence, or otherwise, find the range of possible values of M . (3)

Summer 2022 Q5 — detailed solution

Reaction positivity and parameter range

ALL
9 marks**Set the positions**

Take distances from A. Then C is at 1 m, D is at 3 m, the rod's centre of mass is at 2.5 m, and B is at 5 m.

Find R_C by moments about D

Taking moments about D: $R_C(2) + Mg(2) = 10g(3) + 5g(0.5)$.

Simplify R_C

$2R_C + 2Mg = 30g + 2.5g = 32.5g$, so $R_C = 16.25g - Mg = (16.25 - M)g$.

Find R_D by moments about C

Taking moments about C: $10g(1) + R_D(2) = 5g(1.5) + Mg(4)$.

Simplify R_D

$10g + 2R_D = 7.5g + 4Mg$, so $2R_D = (4M - 2.5)g$ and $R_D = (2M - 1.25)g$.

Use non-negative reactions

For the rod to rest on both supports, both reactions must be non-negative.

Find the range

$R_C \geq 0$ gives $16.25 - M \geq 0$, so $M \leq 16.25$. Also $R_D \geq 0$ gives $2M - 1.25 \geq 0$, so $M \geq 0.625$. Therefore $0.625 \leq M \leq 16.25$.

Final answer (a) $R_C = (16.25 - M)g$. (b) $R_D = (2M - 1.25)g$. (c) $0.625 \leq M \leq 16.25$.

TOPIC 10 | FAMILY 02

Locate the centre of mass of a rod

WHAT THIS FAMILY ASKS

Infer or use the position of a rod's weight, especially for non-uniform rods.

VARIANT ROUTE

V01 Infer centre from reaction data

3 classified questions

25 total marks

Summer 2021 Q7 — complete original question

7.

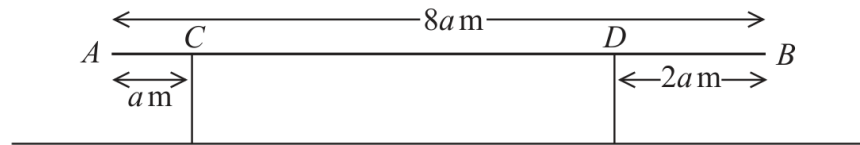


Figure 2

A non-uniform beam AB , of mass 60 kg and length $8a$ metres, rests in equilibrium in a horizontal position on two vertical supports. One support is at C , where $AC = a$ metres and the other support is at D , where $DB = 2a$ metres, as shown in Figure 2.

The magnitude of the normal reaction between the beam and the support at D is three times the magnitude of the normal reaction between the beam and the support at C .

By modelling the beam as a non-uniform rod whose centre of mass is at a distance x metres from A ,

- (a) find an expression for x in terms of a . (5)

A box of mass M kg is placed on the beam at E , where $AE = 2a$ metres.

The beam remains in equilibrium in a horizontal position.

The magnitude of the normal reaction between the beam and the support at C is now equal to the magnitude of the normal reaction between the beam and the support at D .

By modelling the box as a particle,

- (b) find the value of M . (5)

Summer 2021 Q7 — detailed solution

Infer centre-of-mass position from reactions

ALL
10 marks**Set the support positions**

The beam length is $8a$. C is at distance a from A, and D is at distance $6a$ from A because $DB = 2a$.

Name the reactions

Let the reaction at C be R . The reaction at D is $3R$.

Use vertical equilibrium

$R + 3R = 60g$, so $4R = 60g$ and $R = 15g$.

Take moments about A

The beam's weight acts at distance x from A. Hence $60gx = R(a) + 3R(6a)$.

Find x

Substitute $R = 15g$: $60gx = 15ga + 45g(6a) = 285ga$. Thus $x = \frac{285a}{60} = \frac{19a}{4}$.

Use equal reactions with the box

When the box is added, let each support reaction be S . Vertical equilibrium gives $2S = (60 + M)g$.

Take moments about A again

The box is at $AE = 2a$. Moments about A give $60g\left(\frac{19a}{4}\right) + Mg(2a) = S(a) + S(6a) = 7aS$.

Solve for M

Substitute $S = \frac{(60 + M)g}{2}$ and cancel ag : $285 + 2M = \frac{7}{2}(60 + M) = 210 + \frac{7}{2}M$. Hence $75 = \frac{3}{2}M$ and $M = 50$.

Final answer (a) $x = \frac{19a}{4}$. (b) $M = 50$.

October 2021 Q1 — complete original question

1.

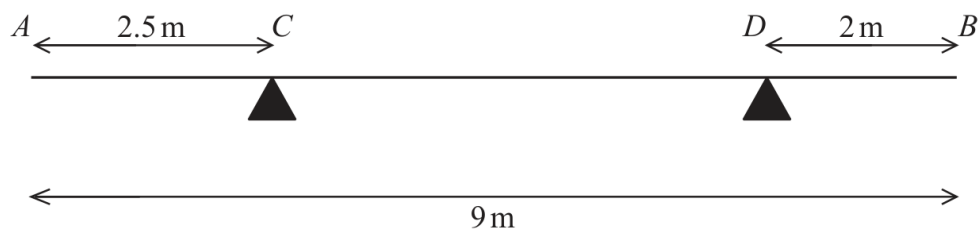


Figure 1

A non-uniform rod AB has length 9 m and mass M kg.

The rod rests in equilibrium in a horizontal position on two supports, one at C where $AC = 2.5$ m and the other at D where $DB = 2$ m, as shown in Figure 1.

The magnitude of the force acting on the rod at D is twice the magnitude of the force acting on the rod at C .

The centre of mass of the rod is d metres from A .

Find the value of d .

(6)

October 2021 Q1 — detailed solution

Infer centre-of-mass position from reactions

ALL
6 marks**Set the support positions**

C is 2.5 m from A. Since $DB = 2$ m and $AB = 9$ m, D is 7 m from A.

Name the reactions

Let the reaction at C be R . The reaction at D is twice this, so it is $2R$.

Use vertical equilibrium

$R + 2R = Mg$, so $3R = Mg$.

Take moments about A

$R(2.5) + 2R(7) = Mgd$.

Substitute $Mg = 3R$

$2.5R + 14R = 3Rd$. Hence $16.5R = 3Rd$.

Find d

Cancel R : $d = \frac{16.5}{3} = 5.5$ m.

Final answer $d = 5.5$ m.

October 2024 Q2 — complete original question

2.

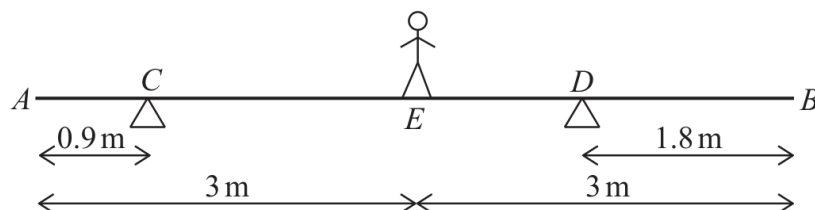


Figure 1

A non-uniform beam AB has length 6 m and mass 50 kg. The beam rests horizontally on two supports at C and D , where $AC = 0.9$ m and $DB = 1.8$ m.

A child of mass 25 kg stands on the beam at E , where $AE = EB = 3$ m, as shown in Figure 1.

The beam is in equilibrium.

The magnitude of the normal reaction between the beam and the support at C is R_C newtons.

The magnitude of the normal reaction between the beam and the support at D is R_D newtons.

The beam is modelled as a rod and the child is modelled as a particle.

The centre of mass of the beam is between C and D and is a distance x metres from D .

Given that $2R_D = 3R_C$

(a) show that $x = 1.38$

(6)

The child remains at E and a block of mass M kg is placed on the beam at B .

The block is modelled as a particle.

Given that the beam is on the point of tilting,

(b) find the value of M .

(3)

October 2024 Q2 — detailed solution

Infer centre-of-mass position from reactions

ALL
9 marks**Find the reactions from vertical equilibrium**

Vertically, $R_C + R_D = 50g + 25g = 75g$. Given $2R_D = 3R_C$, we have $R_D = 1.5R_C$. Therefore $2.5R_C = 75g$, so $R_C = 30g$ and $R_D = 45g$.

Set the distances from D

D is $6 - 1.8 = 4.2$ m from A. The support C is $4.2 - 0.9 = 3.3$ m to the left of D, the child at E is $4.2 - 3 = 1.2$ m to the left of D, and the centre of mass is x m to the left of D.

Take moments about D

Balancing moments about D gives $50gx + 25g(1.2) = R_C(3.3)$.

Show x is 1.38

Substitute $R_C = 30g$: $50x + 30 = 99$, so $50x = 69$ and $x = \frac{69}{50} = 1.38$.

Use the tilting condition with the block

When the block at B is on the point of making the beam tilt, the beam is about to pivot about D and the reaction at C is zero.

Take moments about D for the block

The child and the beam act to the left of D, and the block at B acts 1.8 m to the right of D. Hence $25g(1.2) + 50g(1.38) = Mg(1.8)$.

Find M

$30 + 69 = 1.8M$, so $M = \frac{99}{1.8} = 55$.

Final answer (a) $x = 1.38$. (b) $M = 55$.

TOPIC 10 | FAMILY 03

Impending tilt and loss of support

WHAT THIS FAMILY ASKS

Use the boundary condition that one support force becomes zero and moments balance about the remaining pivot.

VARIANT ROUTE

V01 One support becomes slack

V02 Overhang balanced by a counterweight

V03 Two alternative tipping configurations

6 classified questions

39 total marks

Summer 2026 Q2 — complete original question

2.

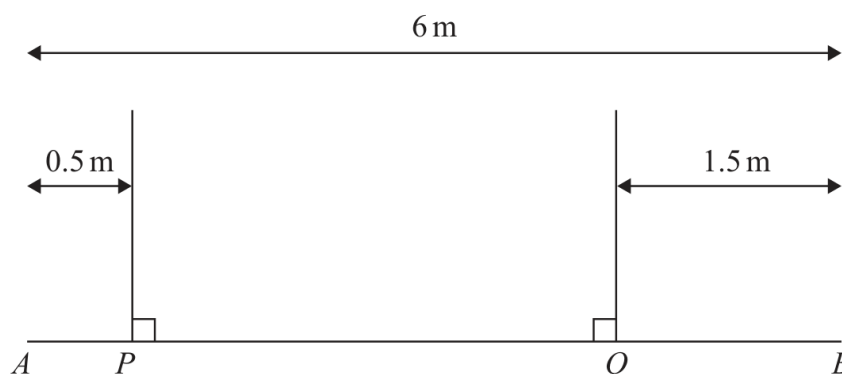


Figure 1

A non-uniform rod AB has mass 20 kg and length 6 m . The rod is held in equilibrium in a horizontal position by two light vertical strings. One string is attached to the rod at P , where $AP = 0.5\text{ m}$ and the other is attached to the rod at Q , where $QB = 1.5\text{ m}$, as shown in Figure 1.

The centre of mass of the rod is 4 m from A .

(a) Find the tension in the string attached to the rod at Q .

(3)

A particle is now attached to the rod at B .

The rod remains in equilibrium in a horizontal position but is on the point of tilting.

The tension in the string attached to the rod at Q is now X newtons.

(b) Find the value of X .

(3)

Summer 2026 Q2 — detailed solution

One support or string becomes inactive

ALL
6 marks**Part (a): put every point on one distance scale**

Measured from A , $AP = 0.5$, $AQ = 4.5$, and the rod's centre of mass is at $AG = 4$ metres. Therefore $PQ = 4$ m and $PG = 3.5$ m.

Take moments about P

The tension at P has zero moment about P . Hence $4T_Q = 20g(3.5)$, so $T_Q = 17.5g = 171.5 \text{ N} = 172 \text{ N}$ to three significant figures.

Part (b): translate point of tilting

The string at P is slack, so its tension is zero and Q is the only active support. The attached particle acts at B , while the rod's weight acts 2 m from B .

Eliminate the attached-particle weight

Taking moments about B gives $1.5X = 20g(2)$. Thus $X = 80g/3 = 261.333 \dots \text{ N} = 261 \text{ N}$ to three significant figures.

Final answer Part (a): $T_Q = 172 \text{ N}$. Part (b): $X = 261 \text{ N}$.

January 2022 Q3 — complete original question

3.

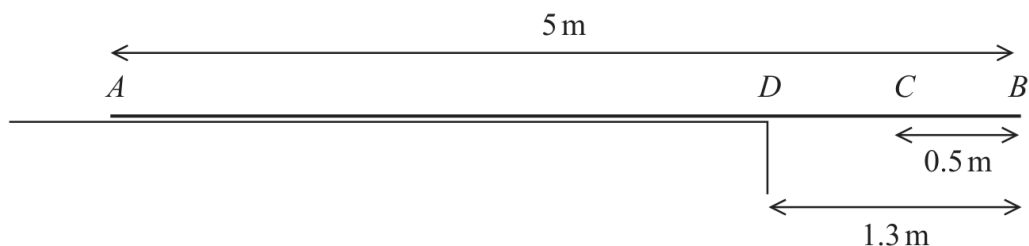


Figure 2

A beam $ADCB$ has length 5 m . The beam lies on a horizontal step with the end A on the step and the end B projecting over the edge of the step. The edge of the step is at the point D where $DB = 1.3\text{ m}$, as shown in Figure 2.

When a small boy of mass 30 kg stands on the beam at C , where $CB = 0.5\text{ m}$, the beam is on the point of tilting.

The boy is modelled as a particle and the beam is modelled as a uniform rod.

(a) Find the mass of the beam.

(3)

A block of mass $X\text{ kg}$ is now placed on the beam at A .

The block is modelled as a particle.

(b) Find the smallest value of X that will enable the boy to stand on the beam at B without the beam tilting.

(3)

(c) State how you have used the modelling assumption that the block is a particle in your calculations.

(1)

January 2022 Q3 — detailed solution

Overhang and counterweight

ALL
7 marks**Set the distances from D**

The beam length is 5 m and $DB = 1.3$ m, so D is 3.7 m from A. The centre of mass of the uniform beam is at its midpoint, so it is 1.2 m to the left of D. Point C is 0.8 m to the right of D.

Use the first tilting condition

When the boy stands at C, the beam is about to tilt about D. Let the mass of the beam be M kg.

Take moments about D

$Mg(1.2) = 30g(0.8)$. Hence $1.2M = 24$ and $M = 20$ kg.

Set up the second tilting condition

With the boy at B, his moment about D is $30g(1.3)$. The block at A is 3.7 m to the left of D, and the beam's weight acts 1.2 m to the left of D.

Take moments about D again

$Xg(3.7) + 20g(1.2) = 30g(1.3)$.

Find X

$3.7X + 24 = 39$, so $3.7X = 15$ and $X = \frac{150}{37} = 4.05 \dots$ kg.

Use the particle model

The block is modelled as a particle, so its mass is treated as acting at one point, A.

Final answer (a) beam mass = 20 kg. (b) $X = \frac{150}{37}$ kg ≈ 4.05 kg. (c) The block's mass is treated as concentrated at A.

Summer 2024 Q4 — complete original question

4.

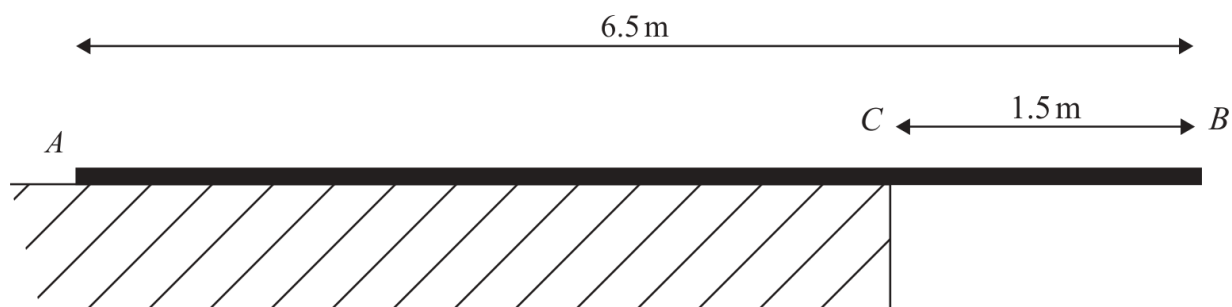


Figure 2

A non-uniform rod AB has length 6.5 m and mass 1.2 kg . The centre of mass of the rod is 3 m from A . The rod rests on a horizontal step and overhangs the end of the step C by 1.5 m , as shown in Figure 2.

The rod is perpendicular to the edge of the step.

A particle of mass 4 kg is placed on the rod at B and another particle, whose mass is $M\text{ kg}$, is placed on the rod at D , where $AD = 0.5\text{ m}$.

The rod remains in equilibrium in a horizontal position.

(a) Find the smallest possible value of M .

(3)

The particle at B **and** the particle at D are now **removed**.

A new particle is placed on the rod at the point E , where $EB = 0.9\text{ m}$.

The rod remains in equilibrium in a horizontal position but is on the point of tilting about C .

(b) Find the magnitude of the force acting on the rod at C .

(3)

Summer 2024 Q4 — detailed solution

Overhang and counterweight

ALL
6 marks**Set the positions relative to C**

The rod is 6.5 m long and overhangs C by 1.5 m, so C is 5.0 m from A. The rod centre of mass is 3.0 m from A, so it is 2.0 m to the left of C. Point D is 0.5 m from A, so it is 4.5 m to the left of C.

Use the limiting balance for smallest M

For the smallest M , the rod is just prevented from tilting clockwise about C. Taking moments about C: $Mg(4.5) + 1.2g(2) = 4g(1.5)$.

Find M

$4.5M + 2.4 = 6$, so $4.5M = 3.6$ and $M = 0.8$.

Place the new particle E

Now the old particles are removed. Since $EB = 0.9$ m and $AB = 6.5$ m, E is 5.6 m from A, so E is 0.6 m to the right of C.

Find the new particle mass from moments

At tilting about C, moments about C give $Xg(0.6) = 1.2g(2)$, so $X = 4$.

Find the force at C

The upward force at C balances the rod and the new particle: $R_C = 1.2g + 4g = 5.2g$ N, which is about 51 N.

Final answer (a) $M = 0.8$. (b) $R_C = 5.2g$ N ≈ 51 N.

January 2021 Q4 — complete original question

4.

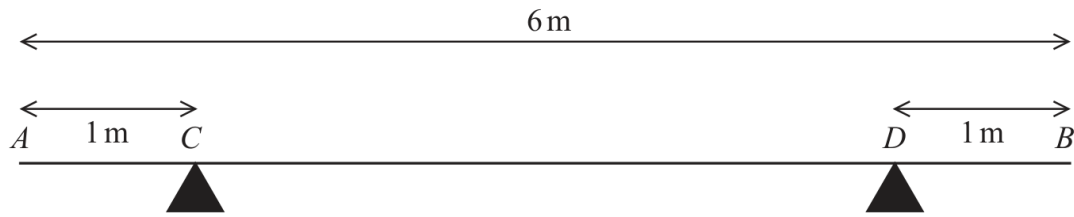


Figure 2

A metal girder AB has weight W newtons and length 6 m. The girder rests in a horizontal position on two supports C and D where $AC = DB = 1$ m, as shown in Figure 2.

When a force of magnitude 900 N is applied vertically upwards to the girder at A , the girder is about to tilt about D .

When a force of magnitude 1500 N is applied vertically upwards to the girder at B , the girder is about to tilt about C .

The girder is modelled as a non-uniform rod whose centre of mass is a distance x metres from A .

Find the value of x .

(6)

January 2021 Q4 — detailed solution

Two possible tipping axes

ALL
6 marks**Set support positions**

The girder has length 6 m, with C 1 m from A and D 1 m from B. So D is 5 m from A.

Use tilting about D

When the 900 N upward force is applied at A, the girder is about to tilt about D. Taking moments about D gives $900(5) = W(5 - x)$.

Use tilting about C

When the 1500 N upward force is applied at B, the girder is about to tilt about C. Taking moments about C gives $1500(5) = W(x - 1)$.

Solve the equations

The equations are $W(5 - x) = 4500$ and $W(x - 1) = 7500$. Add them: $W[(5 - x) + (x - 1)] = 12000$, so $4W = 12000$ and $W = 3000$.

Find x

Use $W(x - 1) = 7500$: $3000(x - 1) = 7500$, so $x - 1 = 2.5$ and $x = 3.5$ m.

Final answer $x = 3.5$ m.

January 2023 Q4 — complete original question

4.

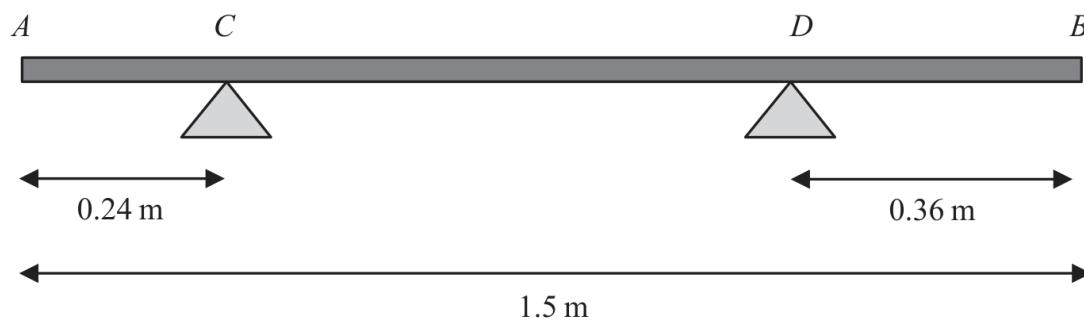


Figure 1

A branch AB , of length 1.5 m , rests horizontally in equilibrium on two supports.

The two supports are at the points C and D , where $AC = 0.24\text{ m}$ and $DB = 0.36\text{ m}$, as shown in Figure 1.

When a force of 150 N is applied vertically upwards at B , the branch is on the point of tilting about C .

When a force of 225 N is applied vertically downwards at B , the branch is on the point of tilting about D .

The branch is modelled as a non-uniform rod AB of weight W newtons.

The distance from the point C to the centre of mass of the rod is x metres.

Use the model to find

- (i) the value of W
- (ii) the value of x

(8)

January 2023 Q4 — detailed solution

Two possible tipping axes

ALL
8 marks**Set the distances**

The branch has length 1.5 m. Since $AC = 0.24$ m and $DB = 0.36$ m, $CB = 1.26$ m and $CD = 0.90$ m. The centre of mass is x m from C.

Use the first tilting condition

With the 150 N upward force at B, the branch is on the point of tilting about C, so the support at D has zero reaction.

Take moments about C

The upward force at B balances the weight moment: $150(1.26) = Wx$. Thus $Wx = 189$.

Use the second tilting condition

With the 225 N downward force at B, the branch is on the point of tilting about D, so the support at C has zero reaction.

Take moments about D

The distance from D to the centre of mass is $0.90 - x$. Hence $225(0.36) = W(0.90 - x)$.

Solve the simultaneous equations

From the second equation, $81 = 0.90W - Wx$. Since $Wx = 189$, $81 = 0.90W - 189$, so $0.90W = 270$ and $W = 300$ N.

Find x

Using $Wx = 189$, $300x = 189$, so $x = 0.63$ m.

Final answer (i) $W = 300$ N. (ii) $x = 0.63$ m.

Summer 2025 Q5 — complete original question

5.

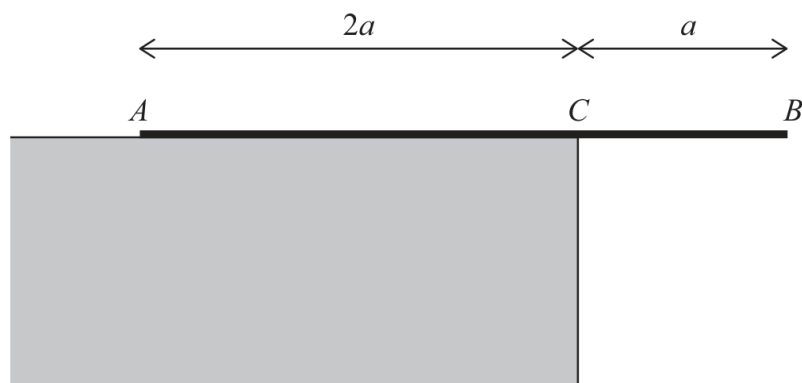


Figure 3

A non-uniform rod AB of length $3a$ rests in equilibrium on a horizontal ledge and overhangs the edge of the ledge at C .

The point C is such that $AC = 2a$ and $CB = a$, as shown in Figure 3.

The rod has weight W .

The distance of the centre of mass of the rod from A is d .

The rod is perpendicular to the edge of the ledge.

When a force of magnitude P , **acting vertically upwards**, is applied to the rod at B , the rod is on the point of tilting about A .

When the force applied at B is replaced by a force of magnitude $1.25P$, **acting vertically downwards** at B , the rod is on the point of tilting about C .

Find d in terms of a .

(6)

Summer 2025 Q5 — detailed solution

Two possible tipping axes

ALL
6 marks**Setup Diagrams - Scenario 1 - Tilting about A**

Taking moments about A:

$$\sum M_A = 0$$

Clockwise moments = Anticlockwise moments

$$P \times 3a = W \times d \quad 3aP = Wd \quad \dots(1)$$

Setup Diagrams - Scenario 2 - Tilting about C

Taking moments about C:

$$\sum M_C = 0$$

Distance from C to COM = $(2a - d)$ Distance from C to B = a

$$1.25P \times a = W \times (2a - d) \quad 1.25Pa = W(2a - d) \quad 1.25Pa = 2aW - Wd \quad \dots(2)$$

Setup Diagrams - Substitute equation (1) into equation (2)From (1): $Wd = 3aP$

$$1.25Pa = 2aW - 3aP \quad 1.25Pa + 3aP = 2aW \quad 4.25Pa = 2aW \quad P = \frac{2aW}{4.25a} \quad P = \frac{2W}{4.25}$$

Substitute back into (1):

$$3a \times \frac{2W}{4.25} = Wd \quad \frac{6aW}{4.25} = Wd \quad d = \frac{6a}{4.25} \quad d = \frac{6a}{17/4} \quad d = \frac{24a}{17} \quad d = 1.41a \quad (3 \text{ sf})$$

Or in exact form: $d = \frac{24a}{17}$

Final answer $d = \frac{24a}{17} = 1.41a.$